

Photonext Viewer - Hololens

User guide

1 Introduction

The PhotoNext project consists of obtaining and processing the displacement and temperature information coming from the sensors of an airplane, so that its status can be visualized simply and immediately through a heat mapped 3D model.

The Hololens application is divided in three main parts: Configuration phase [2], Sensors positioning phase [3] and Monitoring phase [4]. Each of these is described in the following chapters.

2 Configuration phase

To improve the user-friendliness of the software, a *config* file with some information is stored in the `UserFolders\LocalAppData\Photonext-Viewer-Hololens\LocalState` folder.

Since the viewer can be used with different airplanes, more 3d models can be added in the `UserFolders\LocalAppData\Photonext-Viewer-Hololens\LocalState\Models` folder of the program.

2.1 Config file

The *config* file is structured as follows:

- **Connection Type (Local - Database - Past)**

Local - For a direct connection with the middleware through the use of the TCP protocol

Database - For a direct connection to the database

Past - For a direct connection to the database, but for streaming an old collection

- **IP**

It must contain the IP address of the database or middleware

- **Port**

It must contain the port of the database or middleware

- **Username**

The username in case of database/past selection

- **Password**

The password in case of database/past selection

- **Database Name**

The name of the database in case of database/past selection

- **Collection Name**

The name of the collection in case of database/past selection

2.2 Adding of a new 3d model

In case a new 3d model is added in the `\...\LocalState\Models` folder, to work correctly it *must* be in `.obj` extension to work.

2.3 Configuration Panel

The first panel shown when opening the application is the configuration panel: through it, it is possible to import a new 3D model (previously stored in the `\...\LocalState\Models` folder) which, during the monitoring phase, will be used to display the heat map in real time. If the import fails, the application will run with a default model. Subsequently, the user can choose whether to create a new sensor configuration or to import a previously saved one (with the name of the save, name of the model used and date of creation). The selection of a saved configuration will skip the Sensor positioning phase and go directly to the Monitoring phase described in Section 4.

3 Sensors positioning phase

3.1 Sensors configuration

After selecting the *new* button in the configuration panel, you will move on to the new phase, divided into two main parts:

- On the right side it is possible to view the 3D model chosen before or, if no one has been chosen, the default one. It does not follow the user's movements and will remain statically in its position. It can be grabbed with one or two hands to move and resize it.
- On the left side is the panel showing the sensors acquired from the database, represented by a button with a label. There is also a physical object, which can be grabbed and moved until it collides with the model. The panel follows the user's sight and can be grabbed by the label to move it to a position of greater liking. For each sensor, it is possible to set the type between *Temperature* and *Displacement* by simply pressing the selection button. It is possible to intuitively understand the type of sensor based on the shape of the object: it will be a cube in the case of temperature sensors or it will be a sphere in the case of displacement sensors. Since the list of sensors can be quite long, only a limited number of sensors are displayed in the panel at a time. To see the others, simply scroll through the list by pointing your finger in the panel between the button and the sensor. The button can be pressed even if all sensors have already been inserted into the model. There are also two other buttons on the panel: Save and Start.

3.2 Save system

Through the save button, you can decide to save the configuration obtained to use it in the future. This is possible even if not all sensors have been placed into the model. The start button works whether the user chooses to save or not: in the latter case, a warning window will appear to confirm the choice. When the object is resized, a sensor is placed in the model. However, it is possible to grab it to remove it; when the fingers (especially the thumb) are close to the sensor to be removed, it will become larger for an easier selection.

After selecting the start button, the position and rotation of the model will remain the same during the monitoring phase.

4 Monitoring phase

In this part it is possible to view the intensity of each sensor previously positioned through a heat map. It will have a different color depending on the type of sensor:

- **Temperature**

The displayed colors are green, yellow and red, in order of intensity.

- **Displacement**

The displayed colors are light blue, dark blue and dark purple, in order of intensity.

In the same window, a panel containing a graph is also present. It shows, for each sensor, the wavelength variation between -1 and +1 nm over time. Through this panel, it is possible:

- Hide or show both the graph and the 3D model
- Decide whether it should remain in the same position or follow the movement of the user's head

Finally, the Quit button just closes the application.